

Complex Number 12

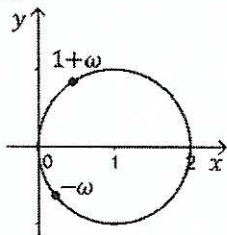
1. If $1, \omega, \omega^2$ are the 3 cube roots of unity:
 - a. Show that $1 + \omega$ is a root of the equation $z^3 - 3z^2 + 3z - 2 = 0$
 - b. Find the integer root of this equation and the third root in terms of ω .
 - c. Graph these 3 roots on an Argand diagram and find the centre and radius of the circle on which they lie.
2. a. If ω is a complex cube root of unity, prove that $1 + \omega + \omega^2 = 0$.
b. Form an equation whose roots are $4, 1 + \omega, 1 + \omega^2$.
3. a. Given that $z = \cos \theta + i \sin \theta$, prove that $z^n + \frac{1}{z^n} = 2 \cos n\theta$
b. Hence, or otherwise, solve $2x^4 + 3x^3 + 5x^2 + 3x + 2 = 0$.
4. The locus of z is represented by $|z - 1| = 1$.
 - a. Sketch the locus
 - b. Show that $|z^2 - z| = |z|$
 - c. Show that $\arg(z - 1) = \frac{2}{3} \arg(z^2 - z)$.
5. a. Find the value of w if i is a root of the equation $z^2 + wz + (1 + i) = 0$
b. Find the real number a and b given that $3 - 2i$ is a root of the equation $z^2 + az + b = 0$.
c. Given that $1 - 2i$ is a root of the equation $z^2 - (3 + i)z + k = 0$, find k and the other root of the equation.
6. a. Solve $z^2 - 2z \cos \theta + 1 = 0$ for z by completing the square
b. Rearranging the equation in part a. given $\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$. Confirm this result for each of the solutions to part a. by substitution.
7. Suppose that $z = \alpha$ is a complex solution to a quadratic equation with real coefficients
 - a. Which other number is also a solution of this quadratic equation?
 - b. Hence prove that the quadratic equation is $z^2 - 2\operatorname{Re}(\alpha)z + \alpha\bar{\alpha} = 0$
8. Let $(x + iy)^2 = a + ib$, then we have $x^2 - y^2 = a$ and $2xy = b$.
 - a. For the moment, assume that both a and b are positive.
 - i. Sketch the graphs of these two equations on the same number plane.
 - ii. What feature of your sketch indicates that there are two square roots of $a + ib$?
 - b. Investigate how the sketch changes when either a or b or both are negative or zero.
9. a. Show that $\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} = \sin \theta + i \cos \theta$
b. Hence prove that $\left(\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} \right)^n = \cos \left(\frac{n\pi}{2} - n\theta \right) + i \sin \left(\frac{n\pi}{2} - n\theta \right)$ where n is a positive integer
10. a. Solve $\tan 4\theta = 1$ for $0 \leq \theta \leq \pi$
b. Express $\tan 2\theta$ in terms of $\tan \theta$
c. Hence show $\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$
d. Hence show $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$ has roots $\tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, \tan \frac{9\pi}{16}$ and $\tan \frac{13\pi}{16}$
e. Hence evaluate $\tan \frac{\pi}{16} \tan \frac{5\pi}{16} \tan \frac{9\pi}{16} \tan \frac{13\pi}{16}$
f. By solving $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$ another way, show the exact value of $\tan \frac{\pi}{16} - \cot \frac{\pi}{16} = -2 - 2\sqrt{2}$
11. a. Show that $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta}$ where $\cos \theta \neq 0$ and n is a positive integer
b. Hence show that if z is a purely imaginary number, the roots of $(1 + z)^4 + (1 - z)^4 = 0$ are $z = \pm i \tan \frac{\pi}{8}, \pm i \tan \frac{3\pi}{8}$

12. a. Express the roots of the equation $z^5 + 32 = 0$ in modulus/argument form
 b. Hence show that: $z^4 - 2z^3 + 4z^2 - 8z + 16 = \left\{z^2 - \left(4 \cos \frac{\pi}{5}\right)z + 4\right\} \left\{z^2 - \left(4 \cos \frac{3\pi}{5}\right)z + 4\right\}$
 c. Hence find the exact values of $\cos \frac{\pi}{5}$ and $\cos \frac{3\pi}{5}$ in simplest surd form.
13. a. Show that the roots of the equation $z^{10} = 1$ are given by $z = \cos \frac{r\pi}{5} + i \sin \frac{r\pi}{5}$, $r = 0, 1, 2, \dots, 9$
 b. Explain why the equation $\left(\frac{z-1}{z}\right)^{10} = 1$ has only nine roots. Show that the roots of $\left(\frac{z-1}{z}\right)^{10} = 1$ are given by $z = \frac{1}{2} \left(1 + i \cot \frac{r\pi}{10}\right)$, $r = 1, 2, 3, \dots, 9$
14. If α and β are the roots of $x^2 - 2x + 4 = 0$, prove that $\alpha^n - \beta^n = i2^{n+1} \sin \left(\frac{n\pi}{3}\right)$. Hence find the value of $\alpha^9 - \beta^9$.
15. If $w = \cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5}$ where k is an integer and $w \neq 1$.
 a. Show that $w^n + w^{-n} = 2 \cos \frac{2nk\pi}{5}$
 b. Show that $w^5 = 1$
 c. Show that $1 + w + w^2 + w^3 + w^4 = 0$
 d. Hence or otherwise show that $(w + w^{-1})^2 + (w^2 + w^{-2})^2 = 3$
 e. Deduce that $\left(\cos \frac{2k\pi}{5}\right)^2 + \left(\cos \frac{4k\pi}{5}\right)^2 = \frac{3}{4}$
16. a. Solve $z^8 + 1 = 0$. Leave answers in mod-arg form.
 b. Factorise $z^8 + 1$ into real quadratic factors.
 c. Show that $\cos 4\theta = 8 \left(\cos^2 \theta - \cos^2 \frac{\pi}{8}\right) \left(\cos^2 \theta - \cos^2 \frac{3\pi}{8}\right)$
17. Let ω be one of the cube roots of unity.
 a. Find the two possible values of $1 + \omega + \omega^2$.
 b. If ω is one of the non-real cube roots of unity, find the value of $(2 - \omega)(2 - \omega^2)(2 - \omega^4)(2 - \omega^5)$

Answers

1. b. $\alpha = -\omega$;

c.



2. b. $x^3 - 5x^2 + 5x - 4 = 0$ (equivalent)

3. b. $z = -\frac{1}{4} \pm \frac{\sqrt{15}}{4}i$ or $z = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$.

5. a. $w = -1$ b. $a = -6$ and $b = 13$ c. $k = 8 - i$

10. a. $\theta = \frac{\pi}{16}, \frac{5\pi}{16}, \frac{9\pi}{16}, \frac{13\pi}{16}$ b. $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$ e.

$\tan \frac{\pi}{16} \tan \frac{5\pi}{16} \tan \frac{9\pi}{16} \tan \frac{13\pi}{16} = 1$

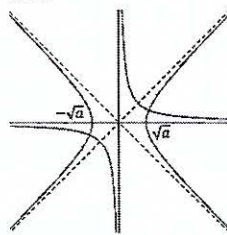
12. a. $2 \operatorname{cis} \frac{2n+1}{5}\pi, n = 0, \pm 1, \pm 2$ c. $\cos \frac{\pi}{5} = \frac{1}{4}(1 + \sqrt{5})$, $\cos \frac{3\pi}{5} = \frac{1}{4}(1 - \sqrt{5})$

14. 0

6. a. $\cos \theta + i \sin \theta$ or $\cos \theta - i \sin \theta$

7. a. $\bar{a}$

8. a. i .



16. a.

$\operatorname{cis} \frac{\pi}{8}, \operatorname{cis} \frac{3\pi}{8}, \operatorname{cis} \frac{5\pi}{8}, \operatorname{cis} \frac{7\pi}{8}, \operatorname{cis} \frac{9\pi}{8}, \operatorname{cis} \frac{11\pi}{8}, \operatorname{cis} \frac{13\pi}{8}, \operatorname{cis} \frac{15\pi}{8}$

b. $z^8 + 1 = (z - \operatorname{cis} \frac{\pi}{8}) \dots (z - \operatorname{cis} \frac{15\pi}{8})$

17. a. $1 + \omega + \omega^2 = 0$ or 3; b. 49